Worksheet 5-7 & 5-12 A
2nd Derivative Test and Implicit Functions #1
AP Calculus Name_____

NO CALCULATORS

x	0	1	4	6	9
$f(x)$	7	12	9	2	5
$f'(x)$	2	0	-3	0	1
$f''(x)$	3	1	0	-2	0

1. Let $f(x)$ be a twice differentiable function with selected values of f, f' and f'' given in the table above. For which x value(s) does $f(x)$ have a relative minimum? Give a reason for your answer.

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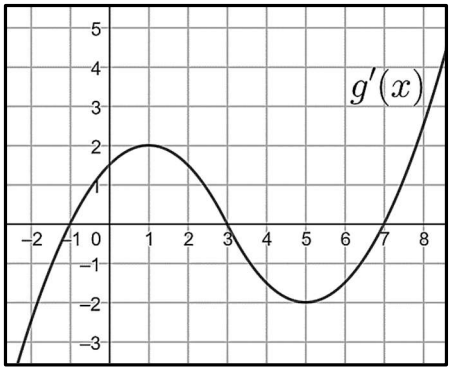
2. Consider the function $y = f(x)$ where $\frac{dy}{dx} = \frac{8-2x}{3y^2}$. It is known that $f(4) = 3$.

A) Find $\frac{d^2y}{dx^2}$.

B) Does $f(x)$ have a relative minimum, relative maximum, or neither at the point (4,3)? Explain your reasoning.



x	-1	0	2	3	7
$h(x)$	-4	3	0	-2	0
$h'(x)$	6	-7	1	0	5

3. Let f be a twice differentiable function where $f'(x) = g'(x) \cdot h(x)$. The graph of $g'(x)$ is shown in the figure above along with a table of $h(x)$ and $h'(x)$ at selected values of x.

For each of the following values of x, determine if the graph of $f(x)$ has a local minimum, local maximum, or neither. If it is not possible to determine if $f(x)$ has a local extremum, write "cannot be determined". Give a reason for your answers.

A) $x = -1$

B) $x = 0$

C) $x = 3$

D) $x = 7$

4. The function $y = k(x)$ is continuous for all real values of x and has the differential equation $\frac{dy}{dx} = \frac{2x+3}{3(y+1)^2}$. It is known that $k(2) = -1$.

A) At which value(s) of x does the graph of k have a vertical tangent line?

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5. For $x > 0$, the function $y = f(x)$ is continuous with $\frac{dy}{dx} = \frac{y}{x} + 1$. It is known that $f(1) = -1$.

A) Write an equation of the line tangent to the graph of f at $x = 1$.

B) Find an expression for $\frac{d^2y}{dx^2}$ in terms of x and y.

C) What is the value of $\frac{d^2y}{dx^2}$ when $x = 1$? Does f have a local minimum, local maximum, or neither at $x = 1$? Give a reason for your answer.

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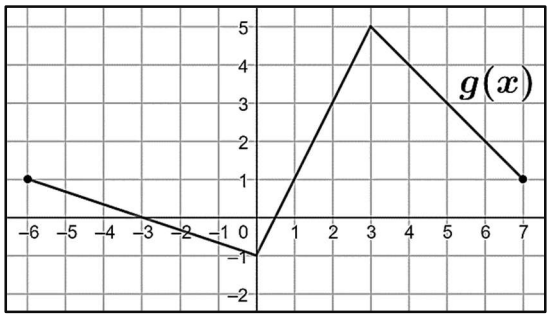
6. Consider the curve defined by $\frac{dy}{dx} = y \sin x$ where $f(2\pi) = 1$. Does f have a local minimum, local maximum, or neither at the given point? Give a reason for your answer.

7. Consider the curve defined by $3x^2 - xy + 2y = -36$ with $\frac{dy}{dx} = \frac{y - 6x}{2 - x}$.

A) Show that there are no points where the line tangent to the curve is vertical.

B) Find the two points where the curve has a horizontal tangent line.

C) Write an equation of the line tangent to the curve when $x = 0$.



x	0	1	2	3	5
$h'(x)$	0	-6	0	2	0
$h''(x)$	4	-5	-3	0	-1

The function $g(x)$ is continuous on the interval $[-6, 7]$ and consists of three line segments as shown in the figure above. The function $h(x)$ is twice differentiable with $h(x) < 0$ for all real values of x. Selected values for h' and h'', the derivatives of h, are included in the table above.

8. Let $f(x)$ be a function where $f'(x) = g(1 - 2x)$. Does f have a local minimum, local maximum, or neither at $x = 2$? Give a reason for your answer.

9. Let $p(x)$ be a function where $p'(x) = g(x^2 + x)$. Does p have a local minimum, local maximum, or neither at $x = -1$? Give a reason for your answer.

10. Does the function h have a relative minimum, relative maximum, or neither at $x = 5$? Explain your reasoning.

11. Does the function h have a relative minimum, relative maximum, or neither at $x = 3$? Explain your reasoning.